

Analog Computation Using Op-Amps

Laboratory Report

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1. Analog Computation

Introduction to Analog Computation

Analog computation uses electronic circuits to simulate differential equations of systems under study. The focus is on linear differential equations of the general form:

$$a_n \frac{d^n x}{dt^n} + a_{n-1} \frac{d^{n-1} x}{dt^{n-1}} + \cdots + a_1 \frac{dx}{dt} + a_0 x = f(t) \quad (1)$$

To simulate this equation, it is rearranged to express the highest derivative in terms of lower-order derivatives and the input function:

$$\frac{d^n x}{dt^n} = -\frac{a_{n-1}}{a_n} \frac{d^{n-1} x}{dt^{n-1}} - \cdots - \frac{a_0}{a_n} x + \frac{1}{a_n} f(t) \quad (2)$$

This form is implemented using operational amplifiers configured as integrators, summers, and gain blocks. The simulation involves:

- Summation Points: Combine scaled inputs.
- Integrators: Obtain lower-order derivatives by successive integrations.
- Gain Blocks: Scale variables according to coefficients.

Amplitude and Time Scaling

- Amplitude Scaling: Adjusts signal levels to prevent saturation and minimize noise. Scale factors relate machine variables (voltages) to problem variables.
- Time Scaling: Changes the simulation speed relative to real time by substituting $t = \sigma\tau$, where σ is the time scaling factor. Adjusting integrator components (resistors and capacitors) alters the time scale.

Ancillary Circuits

- Three-Mode Integrator: Allows setting initial conditions, normal operation, and holding the output constant.
- Quadrature Oscillator: Generates sinusoidal signals with operational amplifiers.
- Voltage Reference: Provides stable voltage levels using operational amplifiers and Zener diodes.
- Chopper Stabilization: Reduces amplifier voltage drift and input current, enhancing low-frequency performance.

Practical limits of IC741: The IC741 requires dual supply rails (e.g. $\pm 12\text{--}15\text{ V}$). The output swing does not reach the rails (expect about 2–3 V margin), and the amplifier has limited slew rate and gain-bandwidth product. Therefore amplitude and time scaling must be chosen to avoid output saturation and slew-induced distortion when implementing the circuits on ELVIS/741 hardware.

Analog computation effectively simulates both linear and nonlinear differential equations using operational amplifiers and associated circuitry. By employing amplitude and time scaling, as well as specialized circuits, analog computers can model complex systems accurately and efficiently.

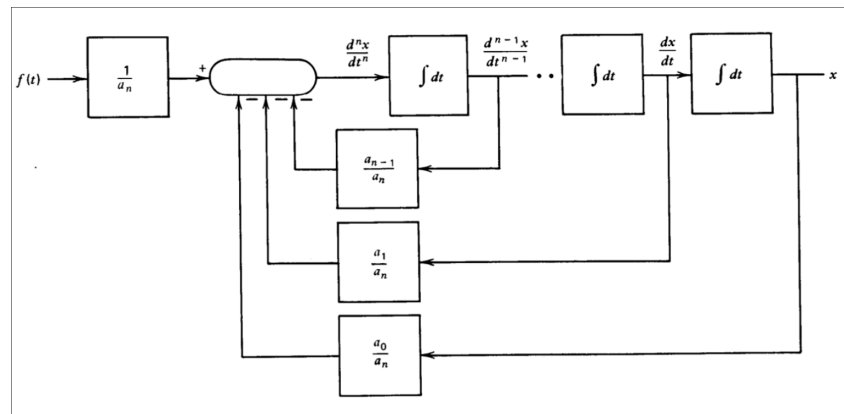


Figure 1: Figure 1: Block diagram of Eqn 1

2. Q1 Linear Combination of 2-signals

Design a circuit to perform a linear combination of two signals $f(t)$ and $g(t)$, i.e.,

$$\alpha \cdot f(t) + \beta \cdot g(t)$$

Solution:

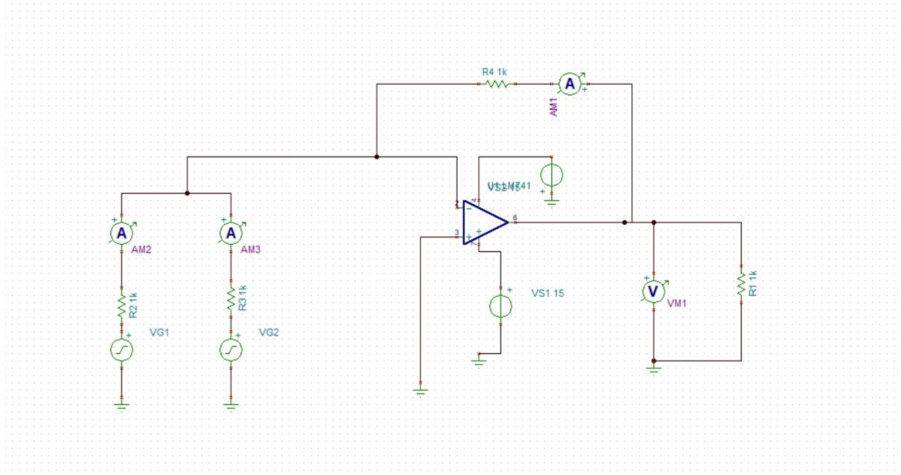


Figure 2: Figure 2: Circuit diagram for the summing amplifier.

The op-amp circuit shown is designed to compute a linear combination of two input signals, $V_f(t)$ and $V_g(t)$, producing an output:

$$V_{out} = \alpha \cdot V_f(t) + \beta \cdot V_g(t)$$

This is achieved using an inverting summing amplifier configuration. In this design, $V_f(t)$ is taken as a sine wave ($\sin(2\pi \cdot 5t)$) and $V_g(t)$ as a cosine wave ($\cos(2\pi \cdot 5t)$), both with a frequency of 5 Hz and an amplitude of 1. The input signals are fed to the inverting terminal through resistors R_2 and R_3 , while the non-inverting terminal is grounded. The resistors scale the input signals, with the scaling factors determined by:

$$\alpha = \frac{R_4}{R_2}, \quad \beta = \frac{R_4}{R_3}$$

where R_4 is the feedback resistor. The negative feedback loop ensures stability and allows the op-amp to sum the currents from the inputs, producing the output voltage:

$$V_{out} = - \left(\frac{R_4}{R_2} \cdot V_f(t) + \frac{R_4}{R_3} \cdot V_g(t) \right)$$

Note on sign: The summing amplifier shown is an *inverting* summing topology, so the algebraic result is negative. If a non-inverting output $+\alpha V_f + \beta V_g$ is required, either use a non-inverting summing topology or append a unity-gain inverter stage to the output.

The circuit diagram demonstrates how the input signals are scaled and summed. The resulting output waveform confirms the desired linear combination, with amplitudes determined by the scaling factors. This design highlights the op-amp's versatility in real-time signal processing, such as audio mixing and control applications.

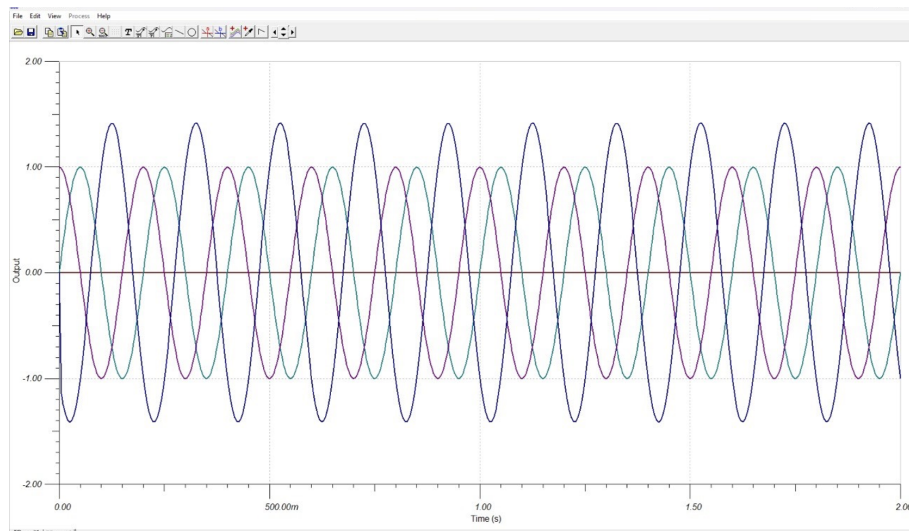


Figure 3: Figure 3: Output Signals

3. Q2 Constraints on the Parameters α and β

To ensure the output voltage $V_o = \alpha f(t) + \beta g(t)$ remains within the operational limits of the ELVIS board and the Op-Amp (IC 741), the following constraints are considered:

1. The ELVIS board imposes the following constraints:

$$V_o \leq 60 \text{ V (peak)}, \quad V_{o,\text{RMS}} \leq 20 \text{ V (RMS)}.$$

2. Given $f(t) = A_1 \sin(\omega_1 t)$ and $g(t) = A_2 \sin(\omega_2 t)$, the output can be expressed as:

$$V_o = \alpha A_1 \sin(\omega_1 t) + \beta A_2 \sin(\omega_2 t).$$

3. The peak constraint translates to:

$$\alpha A_1 + \beta A_2 \leq 60.$$

4. The RMS constraint can be expressed using the formula for the RMS value of the output signal:

$$\text{RMS of } V_o = \sqrt{\left(\frac{\alpha A_1}{\sqrt{2}}\right)^2 + \left(\frac{\beta A_2}{\sqrt{2}}\right)^2}.$$

Simplifying:

$$\sqrt{\frac{(\alpha A_1)^2}{2} + \frac{(\beta A_2)^2}{2}} \leq 20.$$

5. To account for biasing voltage and prevent saturation, we assume the operational limits of the Op-Amp are within $\pm 15 \text{ V}$. To ensure a margin of safety, the design must also comply with the constraints set by this biasing voltage, even though the Op-Amp's high gain (200,000) is not practically significant for this analysis.

These constraints ensure that the output voltage remains within the specified operational range without causing saturation or exceeding the safe limits of the ELVIS board.

4. Q3 Design a Circuit to Implement the Integration of the Function of the Function

Design a circuit to implement the integration of the function:

$$f(x) = x \quad \text{for } 0 < x < 1$$

Evaluate:

$$\int_{-\infty}^x f(t) dt$$

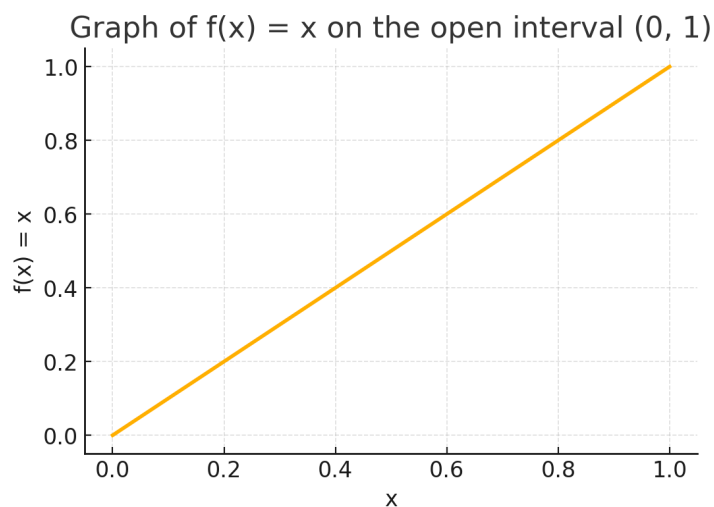


Figure 4: Figure 4: Graph of $f(x) = x$ for $0 \leq x \leq 1$.

Solution:

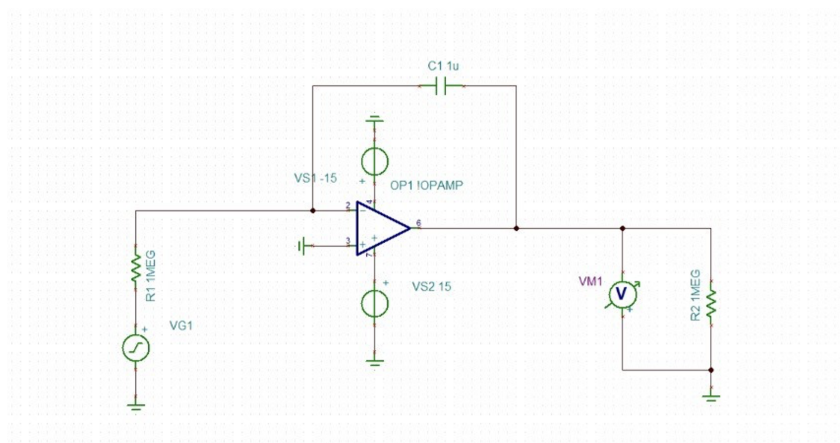


Figure 5: Figure 5: Circuit Diagram

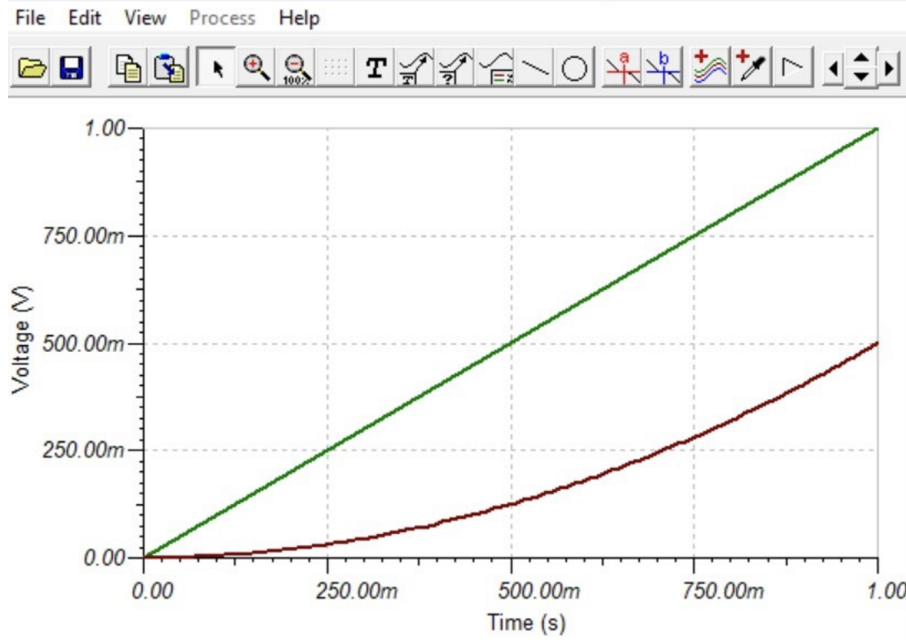


Figure 6: Figure 6: Transient Analysis

The given function $f(x) = x$ is a linear function defined for $0 < x < 1$. The task involves designing a circuit to implement the integration of this function and evaluating the integral

$$\int_{-\infty}^x f(t) dt.$$

Since the function is only defined within the range $0 < x < 1$, the integral is evaluated within this domain.

The integration of $f(x) = x$ over $0 < x < 1$ results in the expression:

$$\int_0^x t dt = \frac{x^2}{2}.$$

This represents the area under the curve of $f(t) = t$ between 0 and x .

To implement this integration in a circuit, an operational amplifier (op-amp) configured as an integrator is used. An op-amp integrator typically includes a resistor at the input and a capacitor in the feedback loop. The input signal $f(t)$ (representing x) is provided as a voltage, and the output voltage corresponds to the integral of the input signal. The mathematical relationship for an ideal op-amp integrator is given by:

$$V_{out}(t) = -\frac{1}{RC} \int V_{in}(t) dt,$$

where R is the resistance, C is the capacitance, $V_{in}(t)$ is the input voltage, and $V_{out}(t)$ is the output voltage representing the integral.

Example component choices (for lab): To test at low frequency (5 Hz) we recommend integrator time constants:

$$R = 100 \text{ k}\Omega, \quad C = 100 \text{ nF} \quad (RC = 0.01 \text{ s}),$$

which gives an integrator corner $\approx 1/(2\pi RC) \approx 15.9 \text{ Hz}$ suitable for 5 Hz inputs. For the ODE coefficients (later in Q6), choose resistor ratios:

$$5 \approx \frac{R_P}{R_Q} \quad \text{with } R_Q = 10 \text{ k}\Omega, R_P = 50 \text{ k}\Omega,$$

$$6 \approx \frac{R_M}{R_N} \quad \text{with } R_N = 10 \text{ k}\Omega, R_M = 60 \text{ k}\Omega.$$

(Use nearest E12 values; include measured actual gains in your lab table.)

5. Q4 Changes in question 3

The spirit realizes that you are succeeding in designing analog computation methods. It destroys your triangular wave generator. How can you perform the integral operation without it?

Solution:

The integration operation can be expressed in terms of Fourier decomposition. Since the function $f(x) = x$ on $[0, 1]$ can be represented using a Fourier sine series:

$$f(x) = \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n\pi} \sin(n\pi x),$$

we compute the integral term by term. The integral of $f(x)$ from $-\infty$ to x is:

$$\int_{-\infty}^x f(t) dt = \int_{-\infty}^x \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n\pi} \sin(n\pi t) dt.$$

This simplifies to:

$$\int_{-\infty}^x f(t) dt = \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n\pi} \int_{-\infty}^x \sin(n\pi t) dt.$$

For each term, the integral of the sine function is:

$$\int \sin(n\pi t) dt = -\frac{\cos(n\pi t)}{n\pi}.$$

Thus, the integral of $f(t)$ becomes:

$$\int_{-\infty}^x f(t) dt = \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{(n\pi)^2} \cos(n\pi x).$$

On truncation and practicality: In practice, the Fourier series must be truncated to the first N harmonics. The truncation error decreases roughly as $O(1/N)$ for this function. Implementing $N = 5$ – 10 harmonics in simulation provides a good approximation. Alternatively, a practical approach is to regenerate the triangular waveform by integrating a square wave using an op-amp integrator.

Implementation Using Sine Waves

1. Generate Sine Wave Inputs

Instead of relying on a triangular wave generator, sine waves corresponding to $\sin(n\pi x)$ for $n = 1, 2, 3, \dots$ are used. These sine waves can be generated using

oscillators. The frequency of each sine wave is proportional to the harmonic number n , with a base frequency f_0 .

2. Integration Using Op-Amps

Each sine wave is processed through an op-amp integrator circuit. The output of the integrator circuit for $\sin(n\pi x)$ is:

$$-\frac{\cos(n\pi x)}{n\pi}.$$

3. Apply Fourier Coefficients

The output of each integrator is scaled by its respective Fourier coefficient $\frac{2(-1)^{n+1}}{n\pi}$. This scaling can be achieved using inverting or non-inverting amplifiers.

4. Summation of Outputs

The scaled outputs of all integrators are combined using a summing amplifier. This summing circuit outputs:

$$\int_{-\infty}^x f(t) dt = \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{(n\pi)^2} \cos(n\pi x).$$

6. Q(5,6) Solving Differential Equation using Op-Amps

The spirit challenges you for a final implementation. You must solve a differential equation using Op-Amps. The spirit presents you with a choice of two differential equations:

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = f(t) \quad (3)$$

with initial conditions:

$$y(0) = 0, \quad \left. \frac{dy}{dt} \right|_{t=0} = 1$$

or

$$\left(\frac{dy}{dt} \right)^2 + 8y = f(t) \quad (4)$$

with initial condition:

$$y(0) = 1$$

Which differential equation will you choose to solve, and why? Provide strong reasoning for your choice.

Solution:

We will choose to solve the first differential equation, as the second one cannot be solved using operational amplifiers (Op-Amps) due to the following reasons:

1. **Nonlinearity:** The term $\left(\frac{dy}{dt}\right)^2$ introduces strong nonlinearity, which Op-Amp methods are not designed to handle. Op-Amps are typically used for solving linear differential equations.
2. **Initial Condition:** The second equation involves an initial-value problem, while Op-Amp methods are better suited for systems with boundary conditions or linear initial-value problems. Handling the squared derivative term further complicates the design.

Thus, the first differential equation is more suitable for implementation using Op-Amps.

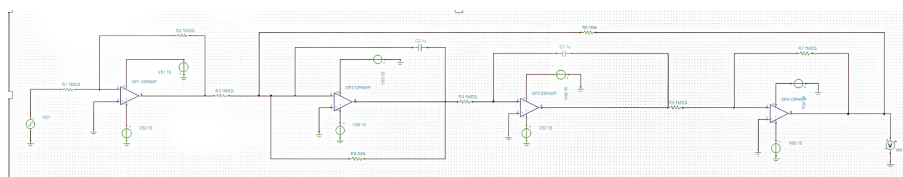


Figure 7: Figure 7: Circuit diagram for solving the first differential equation using Op-Amps.

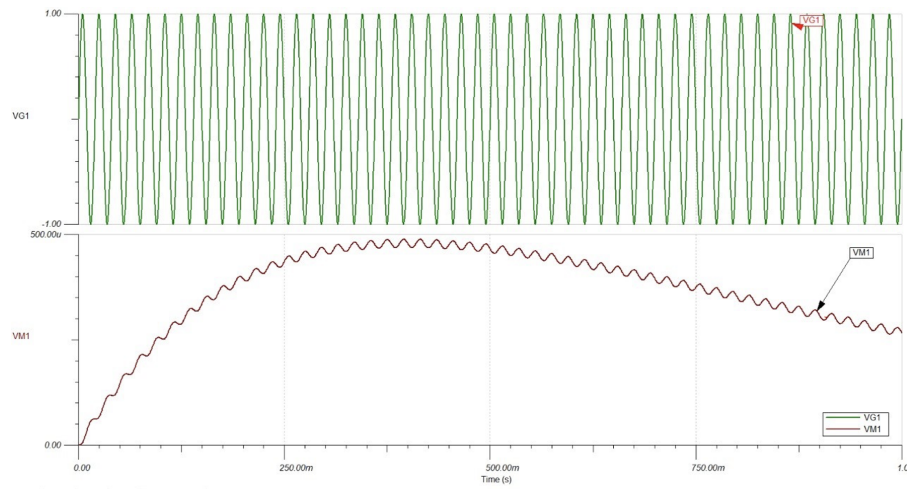


Figure 8: Figure 8: Transient analysis result of the circuit implementation.

Explanation of Circuit Diagram:

This circuit is designed to solve a second-order differential equation using op-amps configured as inverters, integrators and summers. The differential equation it solves is given by:

$$\frac{d^2V}{dt^2} + 5\frac{dV}{dt} + 6V = A \sin(\omega t) \quad (5)$$

The equation can also be written as:

$$\frac{d^2V}{dt^2} = -6V - \left(5\frac{dV}{dt} - A \sin(\omega t) \right) \quad (6)$$

Here, 5 and 6 are coefficients determined by resistor ratios $5 = \frac{R_3}{R_1}$ and $6 = \frac{R_6}{R_4}$. The circuit achieves this solution by breaking the equation into components and solving them using inverters, integrators, multipliers, and summers. The feedback loop ensures that the solution satisfies the equation dynamically.

The circuit consists of four operational amplifiers. The first and second op-amps are configured as integrators, while the third and fourth op-amps perform multiplication and summation. Each op-amp contributes to solving a part of the differential equation.

The first op-amp is an inverter integrator that receives the input $\frac{d^2V}{dt^2}$. It integrates this input and produces $-\frac{dV}{dt}$ as its output. Mathematically, the integrator's operation is defined as:

$$V_{out1} = -\frac{1}{RC} \int \left(\frac{d^2V}{dt^2} \right) dt = -\frac{dV}{dt}. \quad (7)$$

The output $-\frac{dV}{dt}$ from the first op-amp is then fed into the second op-amp, which is also configured as an integrator. The second op-amp integrates the first derivative $-\frac{dV}{dt}$ to produce the signal $V(t)$. The operation of this integrator is expressed as:

$$V_{out2} = -\frac{1}{RC} \int \left(-\frac{dV}{dt} \right) dt = V(t). \quad (8)$$

The third op-amp serves as a multiplier and a summer. Its primary function is to process the first derivative term $-\frac{dV}{dt}$. The circuit multiplies and inverts this term by the coefficient 5. Additionally, the third op-amp adds the forcing function $-V_1$ to the product $5 \cdot \frac{dV}{dt}$. The output of the third op-amp is given by:

$$V_{out3} = 5 \cdot \left(\frac{dV}{dt} \right) - V_1. \quad (9)$$

The fourth op-amp also acts as an inverter, multiplier and a summer. It processes the signal from the second integrator, $-V(t)$, and multiplies it by the coefficient $-Q$, which is determined by the resistor ratio $Q = \frac{R_6}{R_4}$. The output of the fourth op-amp includes

the result of this multiplication and the output of the third op-amp, $(P\frac{dV}{dt} - V_1)$. The mathematical operation of the fourth op-amp is expressed as:

$$V_{out4} = -QV - \left(P\frac{dV}{dt} - V_1\right). \quad (10)$$

This output, which corresponds to the second derivative term $-\frac{d^2V}{dt^2}$, is then fed back into the first op-amp, forming a closed feedback loop.

Setting initial conditions: Use a three-mode integrator or a shorting switch to set the integrator capacitor initial voltage. Sequence: (1) close switch to preset capacitor voltage, (2) open switch to start experiment. To set $\dot{y}(0) = 1$ (in problem units), map the quantity 1 to the corresponding voltage used in your scaling and inject that voltage briefly to the integrator input while presetting the capacitor.

In this feedback loop, the output of the fourth op-amp drives the first op-amp, which continuously integrates the second derivative to compute the first derivative. The second op-amp then integrates the first derivative to produce the solution $V(t)$. The feedback mechanism ensures that the system dynamically adjusts to satisfy the differential equation at all times.

— End of Report —